

1. FCFS Scheduling algorithm

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     int bt[20], wt[20], tat[20];
6
7     printf("Enter number of processes: ");
8     scanf("%d", &n);
9
10    printf("Enter burst time for each process:\n");
11
12    for (i = 0; i < n; i++) {
13        printf("P%d: ", i + 1);
14        scanf("%d", &bt[i]);
15    }
16
17    // Waiting time
18    wt[0] = 0;
19
20    for (i = 1; i < n; i++) {
21        wt[i] = wt[i - 1] + bt[i - 1];
22    }
23
24    // Turnaround time
25    for (i = 0; i < n; i++) {
26        tat[i] = wt[i] + bt[i];
27    }
28
29    printf("\nProcess\tBT\tWT\tTAT\n");
30
31    for (i = 0; i < n; i++) {
32        printf("P%d\t%d\t%d\t%d\n",
33            i + 1, bt[i], wt[i], tat[i]);
34    }
35
36    return 0;
37 }
```

Output:

```
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gedit FCFS.c
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gcc FCFS.c -o FCFS
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ ./FCFS
Enter number of processes: 6
Enter burst time for each process:
P1: 5
P2: 8
P3: 12
P4: 36
P5: 85
P6: 16

Process BT      WT      TAT
P1      5        0        5
P2      8        5       13
P3     12     13     25
P4     36     25     61
P5     85     61    146
P6     16    146    162
```

2. SJF_NONpreemptive

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j;
5     int bt[20], wt[20], tat[20];
6     int temp;
7
8     printf("Enter number of processes: ");
9     scanf("%d", &n);
10
11     printf("Enter burst time:\n");
12
13     for (i = 0; i < n; i++) {
14         printf("P%d: ", i + 1);
15         scanf("%d", &bt[i]);
16     }
17
18     // Sort burst times
19     for (i = 0; i < n - 1; i++) {
20         for (j = i + 1; j < n; j++) {
21             if (bt[i] > bt[j]) {
22                 temp = bt[i];
23                 bt[i] = bt[j];
24                 bt[j] = temp;
25             }
26         }
27     }
28
29     // Waiting time
30     wt[0] = 0;
31
32     for (i = 1; i < n; i++) {
33         wt[i] = wt[i - 1] + bt[i - 1];
34     }
35
36     // Turnaround time
37     for (i = 0; i < n; i++) {
38         tat[i] = wt[i] + bt[i];
39     }
40
41     printf("\nProcess\tBT\tWT\tTAT\n");
42
43     for (i = 0; i < n; i++) {
44         printf("P%d\t%d\t%d\t%d\n",
45             i + 1, bt[i], wt[i], tat[i]);
46     }
47
48     return 0;
49 }
```

Output:

```
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gedit SJF.c
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gcc SJF.c -o SJF
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ ./SJF
Enter number of processes: 5
Enter burst time:
P1: 21
P2: 78
P3: 45
P4: 63
P5: 13

Process BT      WT      TAT
P1      13      0      13
P2      21     13     34
P3      45     34     79
P4      63     79    142
P5      78    142    220
```

3.PRIORITY

```
1 #include <stdio.h>
2
3 int main() {
4
5     int n, i, j;
6     int bt[20], priority[20], wt[20], tat[20];
7     int temp;
8
9     printf("Enter number of processes: ");
10    scanf("%d", &n);
11
12    printf("Enter burst time and priority:\n");
13
14    for (i = 0; i < n; i++) {
15        printf("P%d Burst Time: ", i + 1);
16        scanf("%d", &bt[i]);
17
18        printf("P%d Priority: ", i + 1);
19        scanf("%d", &priority[i]);
20    }
21
22    // Sort according to priority
23    for (i = 0; i < n - 1; i++) {
24        for (j = i + 1; j < n; j++) {
25
26            if (priority[i] > priority[j]) {
27
28                temp = priority[i];
29                priority[i] = priority[j];
30                priority[j] = temp;
31
32                temp = bt[i];
33                bt[i] = bt[j];
34                bt[j] = temp;
35            }
36        }
37    }
38
39    // Waiting time
40    wt[0] = 0;
41
42    for (i = 1; i < n; i++) {
43        wt[i] = wt[i - 1] + bt[i - 1];
44    }
45
46    // Turnaround time
47    for (i = 0; i < n; i++) {
48        tat[i] = wt[i] + bt[i];
49    }
50
51    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");
52
53    for (i = 0; i < n; i++) {
54        printf("P%d\t%d\t%d\t\t%d\t%d\n",
55            i + 1, bt[i], priority[i], wt[i], tat[i]);
56    }
57
58    return 0;
59 }
```

Output:

```
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gedit Priority.c
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gcc Priority.c -o priority
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ ./priority
Enter number of processes: 5
Enter burst time and priority:
P1 Burst Time: 1
P1 Priority: 2
P2 Burst Time: 6
P2 Priority: 3
P3 Burst Time: 8
P3 Priority: 4
P4 Burst Time: 6
P4 Priority: 89
P5 Burst Time: 74
P5 Priority: 56
```

Process	BT	Priority	WT	TAT
P1	1	2	0	1
P2	6	3	1	7
P3	8	4	7	15
P4	74	56	15	89
P5	6	89	89	95

4.ROUND ROBIN

```
1 #include <stdio.h>
2 int main() {
3     int n, i;
4     int bt[20], rem_bt[20];
5     int wt[20], tat[20];
6     int quantum;
7     int time = 0;
8     int done;
9
10    printf("Enter number of processes: ");
11    scanf("%d", &n);
12
13    printf("Enter burst time:\n");
14
15    for (i = 0; i < n; i++) {
16        printf("P%d: ", i + 1);
17        scanf("%d", &bt[i]);
18
19        rem_bt[i] = bt[i];
20        wt[i] = 0;
21    }
22    printf("Enter time quantum: ");
23    scanf("%d", &quantum);
24    while (1) {
25
26        done = 1;
27
28        for (i = 0; i < n; i++) {
29
30            if (rem_bt[i] > 0) {
31
32                done = 0;
33
34                if (rem_bt[i] > quantum) {
35
36                    time = time + quantum;
37                    rem_bt[i] = rem_bt[i] - quantum;
38
39                } else {
40
41                    time = time + rem_bt[i];
42                    wt[i] = time - bt[i];
43                    rem_bt[i] = 0;
44                }
45            }
46        }
47        if (done == 1)
48            break;
49    }
50    for (i = 0; i < n; i++) {
51        tat[i] = wt[i] + bt[i];
52    }
53
54    printf("\nProcess\tBT\tWT\tTAT\n");
55
56    for (i = 0; i < n; i++) {
57        printf("P%d\t%d\t%d\t%d\n",
58            i + 1, bt[i], wt[i], tat[i]);
59    }
60}
```

Output:

```
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gedit Round Robin.c
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gedit RoundRobin.c
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ gcc RoundRobin.c -o roundrobin
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ ./roundrobin
Enter number of processes: 5
Enter burst time:
P1: 4
P2: 9
P3: 63
P4: 12
P5: 87
Enter time quantum: 3

Process BT      WT      TAT
P1      4       12      16
P2      9       22      31
P3     63      85     148
P4     12      34      46
P5     87      88     175
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ |
```